

20.4. IDEAL END CONDITIONS AND EFFECTIVE LENGTH

A column may have two types of end conditions

(i) Position restraint, and (ii) Direction restraint.

An end has a *position restraint* when it is not free to change its position (or is not free to translate), but is free to rotate. The *hinged end* of a column is a typical example of position restraint. In the *direction restraint*, the end of the column is not free to change its direction. It may be free to change its position, or it may not be free to change its position. However, an end of a column may have both the position restraint as well as direction restraint.

Depending upon various combinations of restraints, there may be the following four cases of end conditions (Fig. 20.7) :

Case 1: Both ends of column hinged (Fig. 20.7 a) : *standard case*.

Case 2 : One end of column fixed and the other end free (Fig. 20.7 b).

Case 3 : Both ends of column fixed (Fig. 20.7 c).

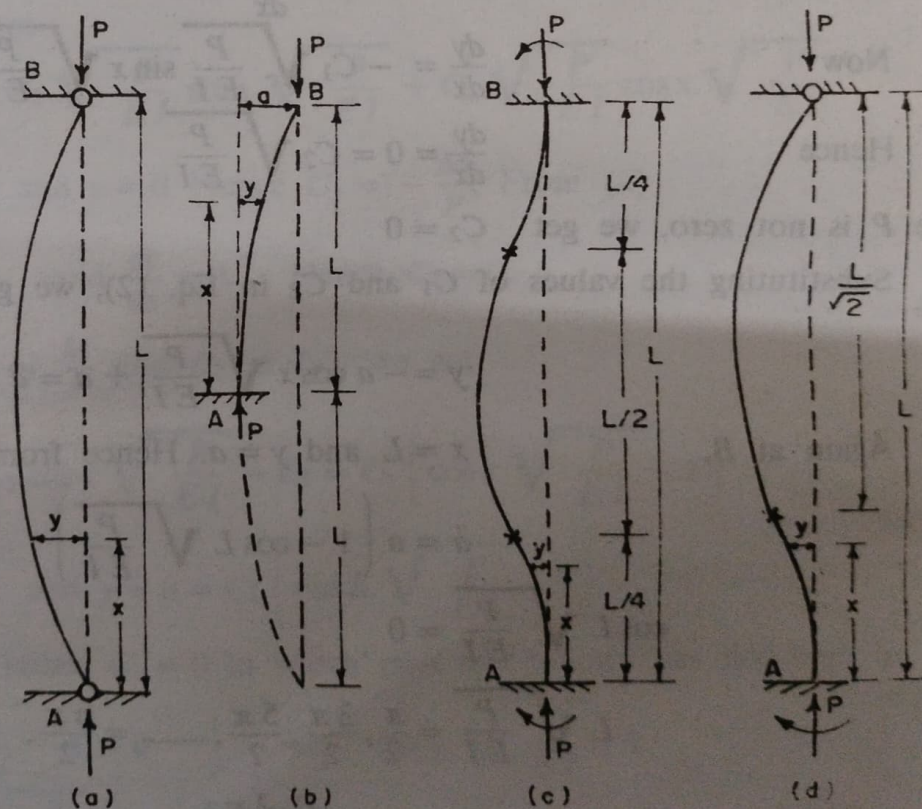


FIG. 20.7. IDEAL END CONDITIONS

Case 4 : One end of column fixed and the other end hinged (Fig. 20.7 d).

Case 1 : Standard Case : Both ends hinged

We have already considered this standard case in the previous article and found that the critical load, or Euler's load is given by

$$P_E = \frac{\pi^2 EI}{L^2} \quad \dots(20.3)$$

Case 2 : One end fixed, other end free (Fig. 20.7 b)

In Fig. 20.7 (b) let end A be direction fixed as well as position fixed and the end B be free. Let a be the deflection at the free end B. At any distance x from the base,

$$M = -P(a - y)$$

Hence the differential equation for the elastic curve is

$$EI \frac{d^2 y}{dx^2} = -M = P(a - y)$$

or

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{Pa}{EI} \quad \dots(1)$$

The solution of the above differential equation is

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + a \quad \dots(2)$$

At A,

$$x = 0 \text{ and } y = 0 \text{ Hence } C_1 = -a$$

Also, at A,

$$x = 0 \text{ and } \frac{dy}{dx} = 0$$

Now

$$\frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}}$$

Hence

$$\frac{dy}{dx} = 0 = C_2 \sqrt{\frac{P}{EI}}$$

Since P is not zero, we get $C_2 = 0$

Substituting the values of C_1 and C_2 in Eq. (2), we get

$$y = -a \cos x \sqrt{\frac{P}{EI}} + a = a \left(1 - \cos x \sqrt{\frac{P}{EI}} \right) \quad \dots(3)$$

Again at B,

$$x = L \text{ and } y = a. \text{ Hence from (3), we get}$$

$$a = a \left(1 - \cos L \sqrt{\frac{P}{EI}} \right)$$

or

$$\cos L \sqrt{\frac{P}{EI}} = 0$$

or

$$L \sqrt{\frac{P}{EI}} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots = \frac{n\pi}{2}, \quad (n = 1, 3, 5, \dots)$$

Hence

$$P_{\sigma} = P_E = \frac{n^2 \pi^2 EI}{4L^2} \quad (n = 1, 3, 5, \dots) \quad \dots(20.6 a)$$

The corresponding buckled shape is $y = a \left(1 - \cos \frac{n \pi x}{2L} \right)$, $n = 1, 3, 5 \dots$... (20.7 a)

The smallest critical load ($n = 1$), is the only load of practical interest.

Hence
$$P_E = \frac{\pi^2 EI}{4L^2} \quad \dots (20.6)$$

The corresponding buckled shape is

$$y = a \left(1 - \cos \frac{\pi x}{2L} \right) \quad \dots (20.7)$$

Case 3 : Both ends fixed (Fig. 20.7 c)

Fig. 20.7 (c) shows a column with both the ends A and B direction fixed as well as position fixed. The column will deflect under the critical load, as shown in Fig. 20.7 (c), with points C and D as points of contraflexure situated at $L/4$ from either end. The column is thus concave for the middle half and convex for a length of $L/4$ from either end. If M is the fixing moment, the differential equation of the elastic curve is given by

$$EI \frac{d^2 y}{dx^2} = M - P \cdot y$$

Rearranging the above equation, we get

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} \cdot y = \frac{M}{EI} \quad \dots (i)$$

The solution of the above differential equation is

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + \frac{M}{P} \quad \dots (ii)$$

Differentiating it, we get

$$\frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}} \quad \dots (iii)$$

At A , $x = 0$ and $y = 0$ Hence $C_1 = -\frac{M}{P}$, From (ii)

At A , $x = 0$ and $\frac{dy}{dx} = 0$. Hence $C_2 = 0$.

Substituting the value of $\frac{M}{P}$ and C_2 in (ii), we get

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} - C_1 = C_1 \left(\cos x \sqrt{\frac{P}{EI}} - 1 \right) \quad \dots (iv)$$

Again at B , $x = L$ and $y = 0 = C_1 \left(\cos L \sqrt{\frac{P}{EI}} - 1 \right)$

This is possible when either $C_1 = 0$ in which case the column has not bent at all, or

$$\cos L \sqrt{\frac{P}{EI}} - 1 = 0 \quad \text{or} \quad \cos L \sqrt{\frac{P}{EI}} = 1$$

This gives $L \sqrt{\frac{P}{EI}} = 0, 2\pi, 4\pi \quad \dots (v)$

Also, at B , $x = L$ and $\frac{dy}{dx} = 0$. Hence from (iii)

$$0 = -C_1 \sqrt{\frac{P}{EI}} \sin L \sqrt{\frac{P}{EI}}$$

or

$$\sin L \sqrt{\frac{P}{EI}} = 0$$

This gives

$$L \sqrt{\frac{P}{EI}} = 0, \pi, 2\pi, 3\pi$$

...(vi)

The minimum significant value of $L \sqrt{\frac{P}{EI}}$, consistent with both (v) and (vi) is 2π .

Hence
$$L \sqrt{\frac{P}{EI}} = 2\pi$$

which gives

$$P_{cr} = P_E = \frac{4\pi^2 EI}{L^2} \quad \dots(20.8)$$

Case 4. One end fixed, the other end hinged

Fig. 20.7 (d) shows a column which is position and direction fixed at end A and hinged at end B . Since end A is fixed, a bending moment M is induced there; this involves the presence of a force R at right angles to AB to maintain equilibrium. The deflected shape is such that there is a point of contra-flexure at point C , distant $L/\sqrt{2}$ from the hinged end B . The differential equation for the elastic curve is

$$EI \frac{d^2 y}{dx^2} = -P \cdot y + R(L - x)$$

or

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{R}{EI} (L - x) \quad \dots(a)$$

Solution of the above differential equation is

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + \frac{R}{P} (L - x) \quad \dots(b)$$

$$\therefore \frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}} - \frac{R}{P} \quad \dots(c)$$

At A , $x = 0$ and $y = 0$ Hence $C_1 = -\frac{RL}{P}$

Also, at A , $x = 0$ and $\frac{dy}{dx} = 0 = C_2 \sqrt{\frac{P}{EI}} - \frac{R}{P}$

Hence $C_2 = \frac{R}{P} \sqrt{\frac{EI}{P}}$

Substituting the values of C_1 and C_2 in (b), we get

$$y = -\frac{RL}{P} \cos x \sqrt{\frac{P}{EI}} + \frac{R}{P} \sqrt{\frac{EI}{P}} \sin x \sqrt{\frac{P}{EI}} + \frac{R}{P} (L - x)$$

Again, at end B ,

$$x = L \text{ and } y = 0$$

$$0 = -\frac{RL}{P} \cos L \sqrt{\frac{P}{EI}} + \frac{R}{P} \sqrt{\frac{EI}{P}} \sin L \sqrt{\frac{P}{EI}}$$

$$\tan L \sqrt{\frac{P}{EI}} = L \sqrt{\frac{P}{EI}}$$

...(d)

or

Hence the value of $L \sqrt{\frac{P}{EI}}$ has to be such that its tangent is equal to the angle itself. The smallest non-zero value of this angle that satisfies Eq. (d) is

$$L \sqrt{\frac{P}{EI}} = 4.4934 \text{ radians} \quad \text{or} \quad \frac{L^2 P}{EI} = 20.19 \approx 2\pi^2$$

Hence

$$P_{cr} = P_E = \frac{20.19 EI}{L^2} \approx \frac{2\pi^2 EI}{L^2} \quad \dots(20.9)$$

Equivalent (or Effective) length

To summarise, Euler's load (critical load) for each one of the above four cases are as under :

$$\text{Case 1 :} \quad P_E = \frac{\pi^2 EI}{L^2}$$

$$\text{Case 2 :} \quad P_E = \frac{\pi^2 EI}{4L^2}$$

$$\text{Case 3 :} \quad P_E = \frac{4\pi^2 EI}{L^2}$$

$$\text{Case 4 :} \quad P_E \approx \frac{2\pi^2 EI}{L^2}$$

In general, Euler's crippling load can be expressed in the form $P_E = \frac{\pi^2 EI}{CL^2}$, in which C is a constant which takes into account the method of fixing the ends (i.e. whether the ends are fixed or hinged or free). For a column hinged at both the ends $C = 1$; hence this case is called the *standard case*. For case 2, $C = 4$; for case 3, $C = \frac{1}{4}$ and for case 4, $C = \frac{1}{2}$. In place of using actual length L of the column we can use length of column which is *equivalent* to that of a pin jointed column. Thus,

$$P_E = \frac{\pi^2 EI}{L_E^2} = \frac{\pi^2 EI}{l^2} \quad \dots(20.10)$$

where L_E (or l) is the *equivalent length* or *effective length*, which is equal to $L\sqrt{C}$, the values of which are given in Table 20.1.

TABLE 20.1. EQUIVALENT LENGTH FOR COLUMNS WITH IDEAL END CONDITIONS

End conditions	Both end hinged	One end fixed and other free	Both ends fixed	One end fixed and other hinged
Equivalent length l or L_E $= L\sqrt{C}$	L	$2L$	$\frac{L}{2}$	$\frac{L}{\sqrt{2}}$

2. Rankine's Formula : Rankine proposed an empirical formula for columns which covers all cases ranging from very short to very long struts. He proposed the relation

$$\frac{1}{P} = \frac{1}{P_C} + \frac{1}{P_E} \quad \dots(20.14)$$

where

$P_C = f_c \cdot A =$ ultimate load for a strut.

$P_E = \frac{\pi^2 EI}{L^2} =$ Eulerian crippling load for the standard case.

In the above relation, $\frac{1}{P_C}$ is constant for a material. For short column, P_E is very large and hence $\frac{1}{P_E}$ is small in comparison to $\frac{1}{P_C}$, thus making crippling load P approximately equal to P_C . For long columns, P_E is extremely small and hence $\frac{1}{P_E}$ is large as compared to $\frac{1}{P_C}$, thus making the crippling load approximately equal to P_E . Thus the value of P obtained from the above relation covers all cases ranging from short to long columns or struts.

The above relation can be re-arranged as :

$$P = \frac{P_C \cdot P_E}{P_E + P_C} = \frac{P_C}{1 + P_C/P_E} = \frac{f_c \cdot A}{1 + \frac{f_c AL^2}{\pi^2 EI}}$$

or

$$P = \frac{f_c \cdot A}{1 + \left(\frac{f_c}{\pi^2 E} \right) \left(\frac{L}{r} \right)^2} = \frac{f_c \cdot A}{1 + a \left(\frac{L}{r} \right)^2} \quad \dots(20.15)$$

In the above equation, f_c is the crushing stress for the material and a is the Rankine's constant for the material. Though the formula is empirical, the values of constant a is not calculated from the values of f_c , π^2 and E , but is determined experimentally. Table 20.4 gives the values of f_c , and a for the materials commonly used for columns and struts.

TABLE 20.4. RANKINE'S CONSTANTS.

Material	f_c		a (For hinged ends)
	kg/cm^2	N/mm^2	
1. Wrought Iron.	2550	255	1 / 9000
2. Cast Iron	5670	567	1 / 1600
3. Mild Steel	3300	330	1 / 7500
4. Strong Timber	500	50	1 / 750

Eq. 20.15 is the Rankine's formula for the standard case of column hinged at ends. For columns of other end conditions, the values of constants will be changed accordingly. However, since a is a constant for a particular material used as a hinged column, it is not convenient to remember the values of a for other end conditions. It is therefore better to modify the Rankine's formula and remember in the form :

$$P = \frac{f_c \cdot A}{1 + a \left(\frac{L_E}{r} \right)^2} = \frac{f_c \cdot A}{1 + a \left(\frac{l}{r} \right)^2} \quad \dots(20.16)$$

in which a is the Rankine's constant for a particular material and for the standard case of column, and L_E or l is the equivalent length of the column. Since equivalent length is generally known, it can be substituted in the above formula to get the Rankine's formula for a particular case of end conditions. *It should be noted that Eqs. 20.15 and 20.16 do not include the factor of safety.*